Multivariate Gaussians

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CS 215, Data Analysis and Interpretation

Univariate Gaussians

- We have seen univariate Gaussian random variables so far.
- Their density is given as follows, where the parameters are the mean μ and standard deviation σ:

$$\boxed{f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}}$$

Multivariate Gaussians

- What if **X** is a multivariate random variable, that is **X** is a vector with d elements (or a vector in d dimensions)
- Then we have **X** = $(X_1, X_2, ..., X_d)$
- In this case, the mean is now a vector given by $\boldsymbol{\mu} = (\mu_1, \mu_2, ..., \mu_d)$. Here $\mu_i = E(X_i)$.
- The variance is now replaced by:
  - The variances of each element: $(\sigma_i)^2 = E[(X_i-\mu_i)^2]$
  - The covariance between every pair of elements $E[(X_i-\mu_i)(X_j-\mu_j)]$.
  - These can now be arranged in the form of a covariance matrix **C** of size d x d.
  - The diagonal elements are individual variances: $C_{ii} = (\sigma_i)^2 = E[(X_i-\mu_i)^2]$
  - The off-diagonal elements are $C_{ij} = E[(X_i-\mu_i)(X_j-\mu_j)] = C_{ji}$.
- The specific form of the Gaussian density is given on the next slide.

Multivariate Gaussians

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}} \qquad \boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{C})$$

Vector-valued random variable **X**

Point at which the density is evaluated (also vector-valued)

|**C**| is the determinant of the matrix **C**.
In some books $\mathbf{C}^{-1}$ is replaced by a matrix **Q** called the **precision matrix** or **inverse covariance matrix**

We will derive this formula in detail, but before that we need to build some foundational concepts.

Some conventions

- Scalars are lower-case in normal font, eg: x, a, b
- Vectors are by default column-vectors and are denoted by lower case bold fonts, eg: **x**, **y**.
- A vector-valued random variable is denoted in upper case, bold font, eg: **X**.
- Matrices are denoted by upper case bold font, eg: **X**. We won't be dealing with matrix-valued random variables in this course.

Covariance matrix

- It is a symmetric matrix.
- It can be interpreted as follows:

$$C = E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top] = E(\boldsymbol{X}\boldsymbol{X}^\top) - E(\boldsymbol{X})E(\boldsymbol{X})^\top$$

- Both **X** and $\boldsymbol{\mu}$ are column vectors, i.e. d x 1 vector.
- Here, we have the **outer product** of (**X** - $\boldsymbol{\mu}$) and its transpose which is a 1 x d vector.
- The outer product yields a d x d **matrix** – the same dimensions as that of **C**. (*In contrast, the inner product or the dot product yields a scalar.*)
- Another way to interpret this: the matrix **C** is created by multiplying every element of (**X** - $\boldsymbol{\mu}$) with every element of $(\boldsymbol{X} - \boldsymbol{\mu})^\top$ and thus $C_{ij} = E[(X_i-\mu_i)(X_j-\mu_j)]$
- Throughout these lectures, we will assume that **C** is an invertible matrix (it has strictly non-zero determinant, in fact a strictly positive determinant).

Covariance matrix

- Let us look at the following definition again:

$$C = E[(X - \mu)(X - \mu)^\top]$$

- Consider some vector **u** in d-dimensions.
- Then we have the following property:

$$\begin{aligned} u^\top C u &= u^\top E[(X - \mu)(X - \mu)^\top] u \\ &= E[u^\top (X - \mu)(X - \mu)^\top u] \\ &= E[(u^\top (X - \mu))^2] \geq 0 \end{aligned}$$

Both these are the dot product between **u** and **x**-$\boldsymbol{\mu}$

Covariance matrix

- A matrix **C** for which $\mathbf{u}^T\mathbf{C}\mathbf{u} >= 0$ for any vector **u** is called a **positive semi-definite matrix**.
- A matrix **C** for which $\mathbf{u}^T\mathbf{C}\mathbf{u} > 0$ for any vector $\mathbf{u} \neq \mathbf{0}$ is called a **positive definite matrix**.
- A covariance matrix is always positive semi-definite (as shown on the previous slide).
- But not every positive-semidefinite matrix is necessarily a covariance matrix because it may not be symmetric.
- Note that the covariance matrix of any vector-valued random variable is positive semi-definite (not just for Gaussians).

Some terminology: normal random vectors

- A d-element vector **Z** is called a **standard normal random vector** if all its elements are independently drawn from N(0,1).
- A d-element vector **X** is called a **centered normal random vector** if there exists a matrix **A** of size d x k such that **AZ** has the same distribution as **X** where **Z** is a k-element standard normal random vector.

Given an **x** (sample of **X**), there exists **z** (a sample of standard normal **Z**) for which **x** = **Az**

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{Z}$$

$$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{z}$$

i-th row of **A** (size 1 x k)

$$\implies \forall i \in \{1, 2, ..., d\}, x_i = \boldsymbol{A}_{i,.}\boldsymbol{z} = \sum_{j=1}^{k} A_{ij} z_j$$

Some terminology: normal random vectors

- A d-element vector **X** is called a **normal random vector** if there exists a matrix **A** of size d x k and d-element vector $\boldsymbol{\mu}$ such that **AZ** + $\boldsymbol{\mu}$ has the same distribution as **X** where **Z** is a k-element standard normal random vector.

i-th row of **A** (size 1 x k)

Given an **x** (sample of **X**), there exists **z** (a sample of standard normal **Z**) for which **x** = **Az** + $\boldsymbol{\mu}$.

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{\mu}$$

$$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{\mu}$$

$$\implies \forall i \in \{1, 2, ..., d\}, x_i = \boldsymbol{A}_{i,.}\boldsymbol{z} + \mu_i = \sum_{j=1}^{k} A_{ij} z_j + \mu_i$$

Normal random vectors

$$\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{C}) \iff \exists \boldsymbol{\mu} \in \mathbb{R}^d, \boldsymbol{A} \in \mathbb{R}^{d\times k} \text{ such that } \boldsymbol{X} = \boldsymbol{AZ} + \boldsymbol{\mu} \text{ and } \forall i \in \{1, ..., d\}, Z_i \sim \mathcal{N}(0, 1)$$

- Notice that even though the values in **Z** are independent of each other, the different values of **X** are **not necessarily** independent of each other. Why?
- We see that E(**X**) = E(**AZ**+$\boldsymbol{\mu}$) = $\boldsymbol{\mu}$. The linearity of expectation formula applies here as well.
- The covariance matrix of **X** is now given as follows:

$$E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top] = E[\boldsymbol{AZ}(\boldsymbol{AZ})^\top]$$

$$= E[\boldsymbol{AZZ}^\top \boldsymbol{A}^\top] \;\; = \boldsymbol{A}E[\boldsymbol{ZZ}^\top]\boldsymbol{A}^\top = \boldsymbol{AA}^\top$$

Derivation of multivariate Gaussian distribution: step 1

- The mean and covariance matrix are now established, but we now still need to derive the formula for the multivariate Gaussian density stepwise starting from standard normal random variables:

$$\forall i \in \{1, 2, ..., d\}, z_i \sim \mathcal{N}(0, 1) \implies f_{Z_i}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-z_i^2/2\right)$$

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}} \exp\left(-z_i^2/2\right)$$

Joint density is the product of the marginals due to independence

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}\right)$$

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{\mu}$$
$$E(\boldsymbol{X}) = \boldsymbol{\mu}$$
$$\mathrm{Cov}(\boldsymbol{X}) = \boldsymbol{A}\boldsymbol{A}^\top$$

- We know the PDF of **Z** and we know the relationship between **X** and **Z**.
- We want to derive the PDF of **X** using transformation of random variables.
- We have seen this concept previously for the scalar case.
- We now want to derive it for the vector case, but first we will revise it for the scalar case.

Transformation of random variables: scalar case

$$Y = g(X) \text{ for increasing function } g$$

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

$$\therefore f_Y(y) = f_X(g^{-1}(y))(g^{-1})'(y) = \frac{f_X(x)}{g'(g^{-1}(y))}$$

$$\text{For decreasing function } g$$

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

$$\therefore f_Y(y) = -f_X(g^{-1}(y))(g^{-1})'(y) = -\frac{f_X(x)}{g'(g^{-1}(y))} = \frac{f_X(x)}{|g'(g^{-1}(y))|}$$

Transformation of random variables: vector case

$$g : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

$$\boldsymbol{Y} = g(\boldsymbol{X}) \Longleftrightarrow \boldsymbol{X} = h(\boldsymbol{Y});$$

$$h = g^{-1}; h : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

$$d\boldsymbol{y} = Dg(\boldsymbol{x})d\boldsymbol{x}$$

$$Dg(\boldsymbol{x}) = \frac{\partial g(\boldsymbol{x})}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_d} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \cdots & \frac{\partial g_2}{\partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_d}{\partial x_1} & \frac{\partial g_d}{\partial x_2} & \cdots & \frac{\partial g_d}{\partial x_d} \end{bmatrix}.$$

This is called the Jacobian matrix (size d x d) and it consists of pairwise partial derivatives of elements of **y** w.r.t. elements of **x**

Transformation of random variables: vector case

$$g : \mathbb{R}^d \to \mathbb{R}^d$$

$$\boldsymbol{Y} = g(\boldsymbol{X}) \iff \boldsymbol{X} = h(\boldsymbol{Y});$$

$$h = g^{-1}; h : \mathbb{R}^d \to \mathbb{R}^d$$

$$d\boldsymbol{y} = Dg(\boldsymbol{x})d\boldsymbol{x}$$

$$d\boldsymbol{x} = Dh(\boldsymbol{y})d\boldsymbol{y}$$

$$Dh(\boldsymbol{y}) = D(g^{-1})(\boldsymbol{y}) = (Dg(h(\boldsymbol{y})))^{-1}$$

$$|Dh(\boldsymbol{y})| = |Dg(h(\boldsymbol{y}))|^{-1}$$

$$Dh(\boldsymbol{y}) = \frac{\partial h(\boldsymbol{y})}{\partial \boldsymbol{y}} = \begin{bmatrix} \frac{\partial h_1}{\partial y_1} & \frac{\partial h_1}{\partial y_2} & \cdots & \frac{\partial h_1}{\partial y_d} \\ \frac{\partial h_2}{\partial y_1} & \frac{\partial h_2}{\partial y_2} & \cdots & \frac{\partial h_2}{\partial y_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_d}{\partial y_1} & \frac{\partial h_d}{\partial y_2} & \cdots & \frac{\partial h_d}{\partial y_d} \end{bmatrix}.$$

$$Dh(y) = \big(Dg(h(y))\big)^{-1}$$

$$|Dh(y)| = \frac{1}{|Dg(h(y))|}$$

Transformation of random variables: vector case

$$\boldsymbol{Y} = g(\boldsymbol{X}) \iff \boldsymbol{X} = h(\boldsymbol{Y});$$

$$P(\boldsymbol{Y} \in B) = \int_B f_{\boldsymbol{Y}}(\boldsymbol{y})d\boldsymbol{y}$$

$$P(\boldsymbol{Y} \in B) = P(\boldsymbol{X} \in g^{-1}(B)) = \int_{g^{-1}(B)} f_{\boldsymbol{X}}(\boldsymbol{x})d\boldsymbol{x}$$

$$= \int_B \boxed{f_X(h(\boldsymbol{y}))|Dh(\boldsymbol{y})|} d\boldsymbol{y}$$

$$\therefore f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{X}}(\boldsymbol{x})|Dh(\boldsymbol{y})| = \frac{f_{\boldsymbol{X}}(\boldsymbol{x})}{|Dg(\boldsymbol{x})|}$$

- Obtained by change of variables from **x** to **y**.
- This integrand must equal f_Y(**y**) as this is true for any set B

Derivation of multivariate Gaussian distribution: step 2

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{\mu}$$

$$\boldsymbol{Z} = \boldsymbol{A}^{-1}(\boldsymbol{X} - \boldsymbol{\mu})$$

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{f_{\boldsymbol{Z}}(\boldsymbol{A}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}))}{\left|\frac{\partial \boldsymbol{X}}{\partial \boldsymbol{Z}}\right|} = \frac{f_{\boldsymbol{Z}}(\boldsymbol{A}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}))}{|\boldsymbol{A}|}$$

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}\right)$$

Derivation of multivariate Gaussian distribution: step 2

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{f_{\boldsymbol{Z}}(\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))}{\left|\frac{\partial \boldsymbol{X}}{\partial \boldsymbol{Z}}\right|} \quad = \frac{f_{\boldsymbol{Z}}(\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))}{|\boldsymbol{A}|}$$

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}\right)$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))^\top (\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

$$= \frac{\exp\left(-\frac{1}{2}((\boldsymbol{x}-\boldsymbol{\mu}))^\top (\boldsymbol{A}^{-1})^\top \boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

Derivation of multivariate Gaussian distribution: step 2

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}\right)$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))^\top (\boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}))\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

$$= \frac{\exp\left(-\frac{1}{2}((\boldsymbol{x}-\boldsymbol{\mu}))^\top (\boldsymbol{A}^{-1})^\top \boldsymbol{A}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

$$\boldsymbol{C}^{-1} = (\boldsymbol{A}^{-1})^\top \boldsymbol{A}^{-1} = (\boldsymbol{A}\boldsymbol{A}^\top)^{-1}$$

$$\therefore \boldsymbol{C} = \boldsymbol{A}\boldsymbol{A}^\top; |\boldsymbol{C}| = |\boldsymbol{A}||\boldsymbol{A}^\top| = |\boldsymbol{A}|^2$$

Derivation of multivariate Gaussian distribution: step 2

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}\right)$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}\,|\boldsymbol{A}|}$$

$$\boldsymbol{C}^{-1} = (\boldsymbol{A}^{-1})^\top \boldsymbol{A}^{-1} = (\boldsymbol{A}\boldsymbol{A}^\top)^{-1}$$

$$\therefore \boldsymbol{C} = \boldsymbol{A}\boldsymbol{A}^\top; |\boldsymbol{C}| = |\boldsymbol{A}||\boldsymbol{A}^\top| = |\boldsymbol{A}|^2$$

$$\therefore f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}}$$

Inference from the derivation

- The previous derivation tells us that the linear/affine transformation of a Gaussian r.v. is another Gaussian r.v. with an updated mean and covariance matrix.
- If **X** is a Gaussian with mean $\boldsymbol{\mu}$ and covariance matrix **C**, then **Y** = **AX**+**b** (where **A** is a matrix-valued constant and **b** is a vector-valued constant) has the mean E(**Y**) = **A**$\boldsymbol{\mu}$ + **b** and covariance matrix $\mathbf{ACA}^T$.
- Prove this result as an exercise.
- Use the transformation of random variables formula.
- This result holds in any dimension.

Practical application of the derivation: sampling from a multivariate Gaussian distribution

- How does one generate a sample vector **x** from N($\boldsymbol{\mu}$, **C**)?
- This is required in many applications.
- We express **C** = **AA**^{T} and obtain **A** via a Cholesky decomposition. We are assured of finding a unique **A** given a positive-definite **C**.
- Generate d samples from N(0,1) using the MATLAB function `randn` and combine them to create vector **z**.
- Then compute **x** = **Az** + $\boldsymbol{\mu}$.
- As per our derivation, it is clear that **x** ~ N($\boldsymbol{\mu}$, **C**).
- There exists an inbuilt function in MATLAB called `mvnrnd`, but it is still nice to know how multivariate Gaussian samples can be generated.
- How does `randn` work internally? We will study it separately.

Special versions of the multivariate Gaussian

$$\therefore f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}}$$

- What happens if **C** is a diagonal matrix, i.e. only the elements C_{ii} are non-zero (in fact positive – why?) and the off-diagonals are all zero.
- This type of a Gaussian is called an **anisotropic Gaussian**. In such a case the term in the exponent reduces to the following:

$$\boldsymbol{Q} := \boldsymbol{C}$$
$$\boldsymbol{f} := \boldsymbol{x} - \boldsymbol{\mu}$$

$$\boldsymbol{f}^\top \boldsymbol{Q} \boldsymbol{f} = \sum_{i=1}^{d} [\boldsymbol{f}^\top \boldsymbol{Q}]_i f_i$$

Special versions of the multivariate Gaussian

$$\boldsymbol{Q} := \boldsymbol{C}$$
$$\boldsymbol{f} := \boldsymbol{x} - \boldsymbol{\mu}$$

$$\boldsymbol{f}^\top \boldsymbol{Q} \boldsymbol{f} = \sum_{i=1}^{d} [\boldsymbol{f}^\top \boldsymbol{Q}]_i f_i \quad = \sum_{i=1}^{d} \sum_{j=1}^{d} f_j Q_{ij} f_i$$

$$= \sum_{i=1}^{d} \left(\sum_{j=1, j \neq i}^{d} f_j Q_{ij} f_i + f_i f_i Q_{ii} \right) \quad = \sum_{i=1}^{d} f_i^2 Q_{ii} = \sum_{i=1}^{d} \frac{f_i^2}{C_{ii}}$$

$$\therefore f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2} \sum_{i=1}^{d} \frac{(x_i - \mu_i)^2}{C_{ii}} \right)}{(2\pi)^{d/2} \prod_{i=1}^{d} C_{ii}^{\;1/2}}$$

Special versions of the multivariate Gaussian

- If **C** = b**I** where b is a scalar and **I** is the identity matrix, then the PDF reduces to:

$$f_{\mathbf{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}\sum_{i=1}^{d}\frac{(x_i - \mu_i)^2}{b}\right)}{(2\pi b)^{d/2}}$$

  - This type of a Gaussian is called an **isotropic Gaussian**.

Marginalization

- A d-element vector **X** is called a **normal random vector** if there exists a matrix **A** of size d x k and d-element vector $\boldsymbol{\mu}$ such that **AZ** + $\boldsymbol{\mu}$ has the same distribution as **X** where **Z** is a k-element standard normal random vector.
- Hence we have the following for any *i* from 1 to d:

$$X_i = [\boldsymbol{AZ}]_i + \mu_i$$
$$= \sum_{j=1}^{d} A_{ij} Z_j + \mu_i$$

- For each j, Z_j is a standard normal r.v.
- Scaling of Z_j by A_{ij} yields another Gaussian with an updated variance.
- The sum of the d different independent Gaussians yields another Gaussian using the concept of convolution (easy to show using MGFs).
- Thus the 1D marginal PDF of any multivariate Gaussian is a univariate Gaussian

Eigenvectors of the Covariance Matrix **C**

- In order to study this, we will first brush up some concepts.
  - Concept of matrices as geometric transformations
  - Eigenvectors and eigenvalues of a matrix

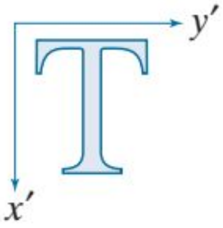
Matrices as Geometric Transformations

- Any matrix **A** of size d x d can be interpreted as a **linear transformation** from R^d to R^d.
- This geometric interpretation of matrices is very useful - a matrix is more than just a square of numbers.
- This is particularly useful given the following definition: “A d-element vector **X** is called a **normal random vector** if there exists a matrix **A** of size d x k and d-element vector $\boldsymbol{\mu}$ such that $\mathbf{AZ} + \boldsymbol{\mu}$ has the same distribution as **X** where **Z** is a k-element standard normal random vector.”

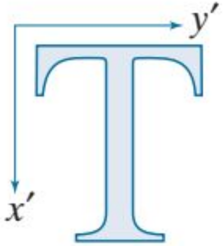
Matrices as Geometric Transformations

- If **A** = b**I** where **I** is the identity matrix, then **A** is called a **(isotropic) scaling transformation**.
- It just scales the original axes (X and Y in 2D; X, Y, Z in 3D; and so on) by the same factors.
- Intuitively, it is transforming a sphere into a concentric larger sphere (if b > 1) or a concentric smaller sphere (if b < 1).

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{aligned} x' &= x \\ y' &= y \end{aligned}$$



$$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{aligned} x' &= c_x x \\ y' &= c_y y \end{aligned}$$



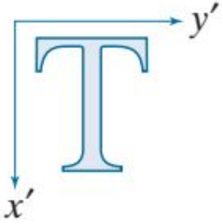
- The illustrations are in 2D here, but they extend to all higher dimensions.
- A *fictitious* third coordinate is introduced here so that even translations can be performed as matrix multiplications. It is called the **homogeneous coordinate**.

Matrices as Geometric Transformations

- If **A** is a diagonal matrix with unequal non-negative diagonal elements, then **A** is called an **anisotropic scaling transformation**.
- It just scales the original axes (X and Y in 2D; X, Y, Z in 3D; and so on) by the different factors.
- Intuitively, it is transforming a sphere into a concentric ellipsoid.

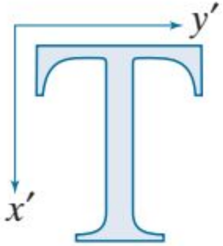
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = x$$
$$y' = y$$



$$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = c_x x$$
$$y' = c_y y$$



Matrices as Geometric Transformations

- A diagonal **A** matrix linearly transforming a standard normal random variable **Z** gives rise to a random variable **X** = **AZ** with a diagonal covariance matrix **C** = $\mathbf{AA}^T$.
- If **A** = b**I**, then **X** has a covariance matrix **C** = $b^2\mathbf{I}$.

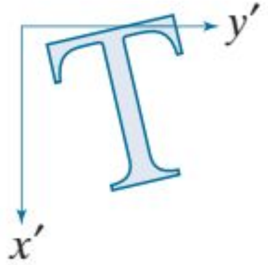
Matrices as Geometric Transformations

- If **A** is an orthonormal matrix, then $\mathbf{A}\mathbf{A}^T = \mathbf{I} = \mathbf{A}^T\mathbf{A}$.
- The determinant of A is either +1 (rotation matrix) or -1 (reflection matrix).
- The covariance matrix of the transformed random variable **X** = **AZ** in this case is the identity matrix.

Rotation (about the origin)

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$



Eigenvectors and eigenvalues

- The vector **v** is said to be the eigenvector of a square matrix **A** of size d x d if there exists a scalar λ (which may be complex-valued even if A is real-valued) for which $\mathbf{Av} = \lambda\mathbf{v}$.
- **Geometric interpretation:** This means that upon the action of **A**, only the magnitude or sign of the vector **v** changes, but its direction does not change (except for the case of reversal of direction if λ=-1).
- A d x d matrix **A** has d eigenvectors corresponding to d eigenvalues. This is of the form $\mathbf{Av_1} = \lambda_1\mathbf{v_1},\ldots, \mathbf{Av_d} = \lambda_d\mathbf{v_d}$.
- This is compactly represented as $\mathbf{AV} = \mathbf{V}\boldsymbol{\Lambda}$ where the columns of **V** are the eigenvectors and $\boldsymbol{\Lambda}$ is the diagonal eigenvalue matrix.
- The eigenvalues may **not** all be unique; some eigenvalues may be **repeated**.

Eigenvectors and eigenvalues of covariance matrix C

- Since the covariance matrix **C** is symmetric, its eigenvectors turn out to form an orthonormal matrix.

Let $A \in \mathbb{R}^{n\times n}$ be symmetric ($A^T = A$). Let $v_i, v_j \in \mathbb{R}^n$ be eigenvectors with eigenvalues λ_i, λ_j, i.e.

$$Av_i = \lambda_i v_i, \qquad Av_j = \lambda_j v_j.$$

Using $Av_j = \lambda_j v_j$:

$$v_i^T A v_j = v_i^T(\lambda_j v_j) = \lambda_j(v_i^T v_j).$$

Using symmetry $A^T = A$ and $Av_i = \lambda_i v_i$:

$$v_i^T A v_j = (A^T v_i)^T v_j = (Av_i)^T v_j = (\lambda_i v_i)^T v_j = \lambda_i(v_i^T v_j).$$

Eigenvectors and eigenvalues of covariance matrix C

Using $Av_j = \lambda_j v_j$:

$$v_i^T A v_j = v_i^T (\lambda_j v_j) = \lambda_j (v_i^T v_j).$$

Using symmetry $A^T = A$ and $Av_i = \lambda_i v_i$:

$$v_i^T A v_j = (A^T v_i)^T v_j = (Av_i)^T v_j = (\lambda_i v_i)^T v_j = \lambda_i (v_i^T v_j).$$

$$\lambda_j (v_i^T v_j) = \lambda_i (v_i^T v_j). \qquad (\lambda_j - \lambda_i)\,(v_i^T v_j) = 0.$$

If $\lambda_i \neq \lambda_j$ then $v_i^T v_j = 0$, i.e. v_i and v_j are orthogonal.

Eigenvectors and eigenvalues of covariance matrix C

- As the eigenvectors of non-equal eigenvalues are perpendicular to each other, it means that any pairs of columns of the eigenvector matrix **V** are orthogonal to each other.
- Hence the eigenvector matrix **V** is an orthonormal matrix. Its determinant may be +1 or -1.
- What if you have repeated eigenvalues?
- Then it that case the two eigenvectors in the previous derivation may not be perpendicular.
- But you can still transform the eigenvector matrix into an orthonormal matrix using something called Gram Schmidt Orthogonalization.
- But that takes us too far into linear algebra and we will skip it.

Eigenvectors and eigenvalues of covariance matrix C

- We have argued previously that **C** is positive semidefinite, that is $\mathbf{u}^T\mathbf{C}\mathbf{u}$ >= 0 for any vector **u.**
- This directly implies that all its eigenvalues are non-negative.
- To see this, observe that:

$$\boldsymbol{C}\boldsymbol{v} = \lambda \boldsymbol{v} \implies \boldsymbol{v}^t \boldsymbol{C} \boldsymbol{v} = \lambda$$

$$\boldsymbol{v}^t \boldsymbol{C} \boldsymbol{v} \geq 0 \implies \lambda \geq 0$$

By our earlier definition of PSD - if is true for any vector **u**, it is also true for eigenvector **v**

Eigenvectors and eigenvalues of covariance matrix C

- We have the following relationship for any matrix C.

$$\boldsymbol{C}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda} \implies \boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$$

- If **C** is a symmetric matrix (and a covariance matrix is always symmetric), **V** is orthonormal so $\mathbf{V}^{-1} = \mathbf{V}^{\top}$. Due to this, we have:

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\top}$$

$$\begin{aligned}\boldsymbol{C}^{-1} &= (\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\top})^{-1} = (\boldsymbol{V}^{\top})^{-1})\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^{-1} \\ &= \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^{\top}\end{aligned}$$

Reformulating the PDF

$$\therefore f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}}$$

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\top$$

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\prod_{i=1}^d \Lambda_{ii}|^{1/2}}$$

$$= \frac{\exp\left(-\frac{1}{2}\left[\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right]^\top \boldsymbol{\Lambda}^{-1}\left[\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right]\right)}{(2\pi)^{d/2}|\prod_{i=1}^d \Lambda_{ii}|^{1/2}}$$

Reformulating the PDF

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\prod_{i=1}^{d}\Lambda_{ii}|^{1/2}}$$

$$= \frac{\exp\left(-\frac{1}{2}\left[\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right]^\top \boldsymbol{\Lambda}^{-1}\left[\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right]\right)}{(2\pi)^{d/2}|\prod_{i=1}^{d}\Lambda_{ii}|^{1/2}}$$

Define $\boldsymbol{f} := \boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})$

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}\boldsymbol{f}^\top\boldsymbol{\Lambda}^{-1}\boldsymbol{f}\right)}{(2\pi)^{d/2}|\prod_{i=1}^{d}\Lambda_{ii}|^{1/2}}$$

The Whitening Transformation

- If **X** has the following PDF, what is the PDF of **H** = $\mathbf{V}^{\top}(\mathbf{X}\text{-}\boldsymbol{\mu})$ where **V** refers to the eigenvectors of **C**, the covariance matrix of **X** and $\boldsymbol{\mu}$ is its mean vector?

$$\therefore f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\top}\boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}}$$

- Note that $\mathbf{X} = \mathbf{V}\mathbf{H}+\boldsymbol{\mu}$. Also $\mathbf{C} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\top}$. The transformed density is:

$$f_{\boldsymbol{H}}(\boldsymbol{h}) = \frac{f_{\boldsymbol{X}}(\boldsymbol{x})}{|\boldsymbol{V}^{\top}|}$$

$$f_{\boldsymbol{H}}(\boldsymbol{h}) = \frac{f_{\boldsymbol{X}}(\boldsymbol{x})}{|\boldsymbol{V}^\top|} \quad = f_{\boldsymbol{X}}(\boldsymbol{V}\boldsymbol{h} + \boldsymbol{\mu})$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{V}\boldsymbol{h} + \boldsymbol{\mu} - \boldsymbol{\mu})^\top \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top(\boldsymbol{V}\boldsymbol{h} + \boldsymbol{\mu} - \boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\prod_{i=1}^d \Lambda_{ii}|^{1/2}}$$

$$= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{V}\boldsymbol{h})^\top \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top(\boldsymbol{V}\boldsymbol{h})\right)}{(2\pi)^{d/2}|\prod_{i=1}^d \Lambda_{ii}|^{1/2}}$$

$$= \frac{\exp\left(-\frac{1}{2}\boldsymbol{h}^\top \boldsymbol{\Lambda}^{-1}\boldsymbol{h}\right)}{(2\pi)^{d/2}|\prod_{i=1}^d \Lambda_{ii}|^{1/2}}$$

The Whitening Transformation

- Consider the transformation of $\mathbf{H} = \mathbf{V}^T(\mathbf{X}-\boldsymbol{\mu})$.
- Even if **X** was a Gaussian with a full covariance matrix, the covariance matrix of **H** is diagonal.
- In other words, the different components of **H** are uncorrelated.
- The aforementioned transformation is called a **whitening** or **decorrelating transformation**.
- The mean value of **H** is **0**, the zero vector.
- The covariance matrix of **H** is $\boldsymbol{\Lambda}$.

Joint and marginal Gaussian PDFs

- If **X** is a normal random vector (or multivariate Gaussian random vector), then we have seen that each of its components is a univariate Gaussian.
- However if two (or more) scalar-valued random variables X and Y are individually Gaussian, it does not mean that their joint distribution is a multivariate Gaussian.
- We show this via a counterexample.
  - Suppose X ~N(0,1), and U is a random variable independent of X such that P(U = +1) = P(U=-1) = 0.5 (Rademacher distribution)
  - Let Y = UX.
  - Then Y = X with probability 0.5 and Y = -X with probability 0.5.
  - Hence $f_Y(y) = 0.5\, f_X(y) + 0.5\, f_X(-y) = 0.5\, f_X(y) + 0.5\, f_X(y) = f_X(y)$ due to symmetry of N(0,1).
  - Thus Y~N(0,1).
  - If (X,Y) were jointly Gaussian, then any linear combination of X and Y should be Gaussian.
  - But X+Y is not Gaussian – X+Y = 0 with probability 0.5 and X+Y=2X with probability 0.5.
  - This clearly violates the basic property of any Gaussian distribution.

Joint and marginal Gaussian PDFs

- Consider two **independent** scalar-valued random variables X and Y that are individually Gaussian distributed.
- Then they are also jointly Gaussian distributed. Their covariance is a diagonal matrix (why?)
- If two random variables X and Y are jointly Gaussian, it does not necessarily mean that they are independent:
    - They are independent only if they are **uncorrelated**, i.e. their covariance matrix is diagonal

Multivariate normals: isocontours

- Consider **X** ~ N($\boldsymbol{\mu}$,**C**).
- Suppose we want to find out all values **x** for which $f_{\mathbf{X}}(\mathbf{x}) = c$ (some constant).

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)}{(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}} = c$$

$$-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) - \log\left((2\pi)^{d/2}|\boldsymbol{C}|^{1/2}\right) = \log c$$

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = -2\log(c(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}) = K(c)$$

Multivariate normals: isocontours

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = -2\log(c(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}) = K(c)$$

$$\forall c, K(c) > 0$$

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\top$$

$$\therefore (\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu}) = K(c)$$

$$\boldsymbol{y} := \boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})$$

$$\boxed{\begin{aligned} \boldsymbol{y}^\top \boldsymbol{\Lambda}^{-1}\boldsymbol{y} &= K(c) \\ \sum_{i=1}^{d} \frac{y_i^2}{K(c)\lambda_i} &= 1 \end{aligned}}$$

This is the equation of a hyper-ellipsoid

Multivariate normals: isocontours

$$\boldsymbol{y}^\top \boldsymbol{\Lambda}^{-1} \boldsymbol{y} = K(c)$$

$$\sum_{i=1}^{d} \frac{y_i^2}{K(c)\lambda_i} = 1$$

- This is the equation of a hyper-ellipsoid
- If **C** = b**I**, then the hyper-ellipsoid reduces to a hyper-sphere as all the eigenvalues of **C** equal b
- If **C** is a diagonal matrix with unequal diagonal values, then the principal axes of the hyper-ellipsoid are aligned with the usual coordinate axes (eg: X,Y,Z in 3D; X,Y in 2D; X_1, X_2,...,X_d in d dimensions)
- Larger eigenvalue → larger spread along that direction
- If **C** is a full covariance matrix, then the hyper-ellipsoid is rotated by **V**, i.e. the principal axes are not aligned with the coordinate axes.
- In the case of diagonal covariances, note that **V** = **I**.

Multivariate normals: isocontours

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = -2\log(c(2\pi)^{d/2}|\boldsymbol{C}|^{1/2}) = K(c)$$

$$\forall c, K(c) > 0$$

$$K(c) = -2\log c - d\log 2\pi - \log|C|$$

$$0 < c \leq p_{max} = f_{\boldsymbol{X}}(\boldsymbol{\mu}) = 2\pi^{-d/2}|\boldsymbol{C}|^{-0.5}$$

$$\log p_{max} = -d/2\log 2\pi - 1/2\log|\boldsymbol{C}|$$

$$K(c) = -2\log c + 2\log p_{max} = -2\log\left(\frac{c}{p_{max}}\right)$$

$$c \leq p_{max} \implies K(c) = -2 \times \text{ some negative number } \implies K(c) > 0$$

Multivariate normals: isocontours

- Why are we interested in studying these isocontours?
- They tell us something about what values of **x** satisfy certain density values.
- They are also useful in classification problems or anomaly detection problems.
- Given a Gaussian distribution N($\boldsymbol{\mu}$,**C**), we say a point **x** is an anomaly w.r.t. this distribution, if $(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})$ > some threshold τ.
- Here the Gaussian distribution is acting as some sort of a signature for "normal" behavior.
- The point **x** is said to be "Abnormal" or "Anomalous" if it doesn't obey the normal behaviour as defined before.

Multivariate normals: isocontours

- In AI/ML and beyond, you will learn about Gaussian mixture models where the PDF is expressed as:

$$f_{\boldsymbol{X}}(\boldsymbol{x}) := \sum_{k=1}^{K} \frac{p_k}{(2\pi)^{d/2}|\boldsymbol{C_k}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_k})^\top \boldsymbol{C_k}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_k})\right)$$

$$\sum_{k=1}^{K} p_k = 1; \forall k, p_k \geq 0$$

- A Gaussian mixture-model is supposed to be "a universal density approximator", i.e. it can approximate any density given sufficiently large K.
- A point **x** will be anomalous w.r.t. the entire mixture model if for every *k* we have: $(\mathbf{x}-\boldsymbol{\mu}_\mathbf{k})^\top \mathbf{C}_\mathbf{k}^{-1}(\mathbf{x}-\boldsymbol{\mu}_\mathbf{k}) > \tau_k$.

Euclidean distance and Mahalanobis distance

- Given two vectors **x** and **y** in R^d, the Euclidean distance between them is given by

$$\|\boldsymbol{x} - \boldsymbol{y}\|_2 = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$$

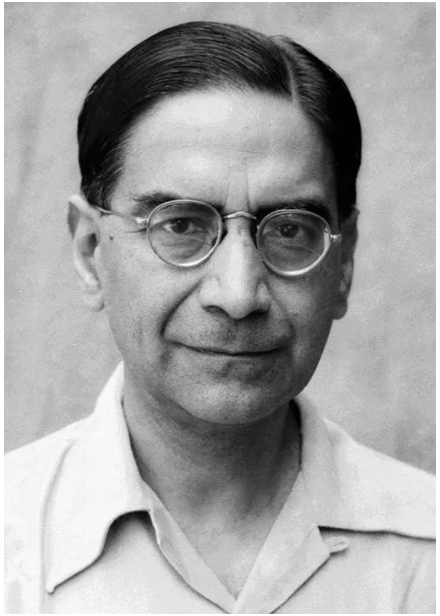
- The Euclidean distance is a very commonly used distance measure (or metric)
- What are the properties that a distance measure d(**x**,**y**) should satisfy?
  - Symmetry: d(**x**,**y**) = d(**y**,**x**)
  - Positivity: d(**x**,**y**) ⩾ 0
  - Idempotence: d(**x**,**x**) = 0
  - Triangle inequality: d(**x**,**y**) ⩽ d(**x**,**z**) + d(**z**,**y**)
- It is easy to verify that the Euclidean distance satisfies these conditions.

Euclidean distance and Mahalanobis distance

- When it comes to evaluating the distance from the mean vector $\boldsymbol{\mu}$ of a Gaussian, we use the term in the exponent given by:

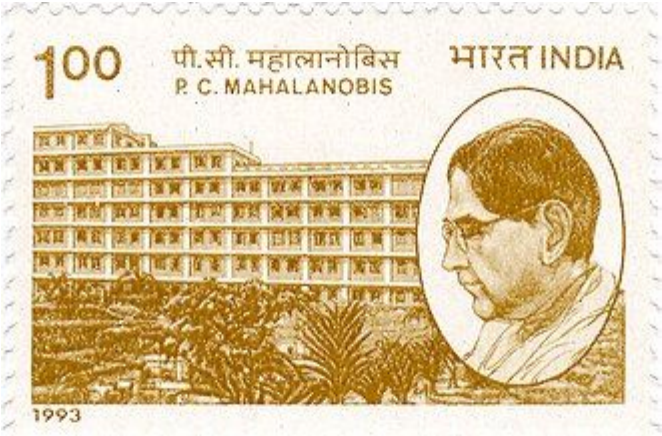
$$\sqrt{(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

- This is called the **Mahalanobis distance**, named after Prasanta Chandra Mahalanobis.
- For an invertible **C**, it satisfies the three conditions for a valid distance measure:
  - Positivity: because **C** is positive semi-definite
  - Symmetry: follows from definition
  - Idempotence: follows from definition
  - Triangle inequality: Proof on next slide



- Legendary Indian statistician: P. C. Mahalanobis (Prasanta Chandra Mahalanobis)
- Founder of the Indian Statistical Institute (ISI) in Kolkata in 1932 – it now has branches in Delhi, Bengaluru, Chennai and Tezpur
- Teacher of other legendary statisticians such as C. R. Rao

https://en.wikipedia.org/wiki/Prasanta_Chandra_Mahalanobis



Euclidean distance and Mahalanobis distance

$$d(\boldsymbol{x}, \boldsymbol{y}) \leq d(\boldsymbol{x}, \boldsymbol{z}) + d(\boldsymbol{z}, \boldsymbol{y})$$

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^\top \boldsymbol{C}^{-1} (\boldsymbol{x} - \boldsymbol{y})}$$

$$\exists \boldsymbol{A}, \text{ such that } \boldsymbol{C} := \boldsymbol{A}\boldsymbol{A}^\top \implies \boldsymbol{C}^{-1} = (\boldsymbol{A}^{-1})^\top \boldsymbol{A}^{-1} = \boldsymbol{B}^\top \boldsymbol{B}$$

This is called the Cholesky decomposition of a matrix: there will be a unique **A** and hence unique **B** (upto a sign) for a positive-definite matrix **C**.

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^\top \boldsymbol{C}^{-1} (\boldsymbol{x} - \boldsymbol{y})} = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^\top \boldsymbol{B}^\top \boldsymbol{B} (\boldsymbol{x} - \boldsymbol{y})} = \|\boldsymbol{B}(\boldsymbol{x} - \boldsymbol{y})\|_2$$

With this reformulation, the Mahalanobis distance is expressed as the Euclidean distance between **Bx** and **By**. So using the triangle inequality for Euclidean distances, we now have:

$$d(\boldsymbol{x}, \boldsymbol{y}) \leq d(\boldsymbol{x}, \boldsymbol{z}) + d(\boldsymbol{z}, \boldsymbol{y})$$

Squared Mahalanobis distance and Euclidean distance

- We make the following observation regarding the squared Mahalanobis distance:

B is obtained from the Cholesky decomposition of **C**

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{B}^\top \boldsymbol{B}(\boldsymbol{x}-\boldsymbol{\mu}) = \|\boldsymbol{B}(\boldsymbol{x}-\boldsymbol{\mu})\|_2^2 = \sum_{i=1}^{d} ([\boldsymbol{B}\boldsymbol{x}]_i - [\boldsymbol{B}\boldsymbol{y}]_i)^2$$

If **x** ~ N(μ, **C**), this suggests that the squared Mahalanobis distance between **x** and μ is equal to the Euclidean distance between vectors **Bx** and **By**.

Squared Mahalanobis distance: chi-squared distribution

- We make the following observation regarding the squared Mahalanobis distance:

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top\right)^{-1}(\boldsymbol{x}-\boldsymbol{\mu})$$

$$= \left(\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)^\top \boldsymbol{\Lambda}^{-1}\left(\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

$$= \boldsymbol{y}^\top\boldsymbol{\Lambda}^{-1}\boldsymbol{y} = \sum_{i=1}^{d}\frac{y_i^2}{\lambda_{ii}}$$

The elements of **x** are correlated, but the elements of **y** are uncorrelated as **V** is a decorrelating transformation. As **y** is (jointly) Gaussian distributed, uncorrelated implies **independent**. So the different terms in the summation are squares of **independent** standard normals (their variance is 1 due to division by λ_{ii}), and hence the square of the Mahalanobis distance obeys a **chi-square distribution with d degrees of freedom**. This is used in hypothesis testing and anomaly detection.

The Mahalanobis distance can be viewed as a multivariate generalization of the Z test statistic: $|x-\mu|/\sigma$

Squared Mahalanobis distance: chi-squared distribution

$$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\boldsymbol{V}\boldsymbol{\Lambda}^{-1}\boldsymbol{V}^\top\right)^{-1}(\boldsymbol{x}-\boldsymbol{\mu})$$

$$= \left(\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)^\top \boldsymbol{\Lambda}^{-1}\left(\boldsymbol{V}^\top(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

$$= \boldsymbol{y}^\top \boldsymbol{\Lambda}^{-1}\boldsymbol{y} = \sum_{i=1}^{d} \frac{y_i^2}{\lambda_{ii}}$$

- In the expression above, note that each λ_{ii} = variance (y_i). Why? Prove it as an exercise.
- Notice how the expression above takes into account that the variance along the different directions is different.
- For example, for a 3D Gaussian with diagonal covariance matrix and variances 1, 10 and 100, the variance along Z is very large. Any sound hypothesis test for such a Gaussian should allow larger distances along the Z axis than along the X or Y axes.

Maximum Likelihood Estimates for Multivariate Gaussians

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$ be i.i.d.

$$\mathbf{x}_i \overset{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$$

$$L(\boldsymbol{\mu}, \mathbf{C}) = \prod_{i=1}^{n} \frac{1}{(2\pi)^{d/2}|\mathbf{C}|^{1/2}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x}_i - \boldsymbol{\mu})\Big)$$

$$\ell(\boldsymbol{\mu}, \mathbf{C}) = -\frac{nd}{2}\log(2\pi) - \frac{n}{2}\log|\mathbf{C}| - \frac{1}{2}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}).$$

$$\frac{\partial}{\partial \boldsymbol{\mu}}\left[(\mathbf{x}_i - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x}_i - \boldsymbol{\mu})\right] = -2\,\mathbf{C}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}).$$

$$\frac{\partial \ell}{\partial \boldsymbol{\mu}} = -\frac{1}{2}\sum_{i=1}^{n}(-2)\,\mathbf{C}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) = \mathbf{C}^{-1}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu}).$$

$$\mathbf{C}^{-1}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu}) = \mathbf{0} \quad \Longrightarrow \quad \sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu}) = \mathbf{0}.$$

$$\boxed{\widehat{\boldsymbol{\mu}} = \overline{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i}.$$

$$\mathbf{Q} := \mathbf{C}^{-1}.$$

$$\ell(\boldsymbol{\mu}, \mathbf{Q}) = \frac{n}{2} \log |\mathbf{Q}| - \frac{1}{2} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})^\top \mathbf{Q} (\mathbf{x}_i - \boldsymbol{\mu}).$$

$$\frac{\partial}{\partial \mathbf{Q}} \log |\mathbf{Q}| = (\mathbf{Q}^{-1})^\top = \mathbf{Q}^{-1}.$$

$$-\frac{1}{2} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top = -\frac{1}{2} S(\boldsymbol{\mu}).$$

$q = \mathbf{a}^\top \mathbf{Q} \mathbf{a} = \sum_{jk} a_j Q_{jk} a_k,\ \partial q / \partial Q_{jk} = a_j a_k.)$

$$\frac{\partial \ell}{\partial \mathbf{Q}} = \frac{n}{2} \mathbf{Q}^{-1} - \frac{1}{2} S(\boldsymbol{\mu}).$$

$$\boldsymbol{S}(\boldsymbol{\mu}) := \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^\top$$

$$\boxed{\mathbf{C} = \frac{1}{n} S(\boldsymbol{\mu})}.$$

$$\boxed{\widehat{\mathbf{C}} = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}})(\mathbf{x}_i - \widehat{\boldsymbol{\mu}})^\top}.$$

The MLE estimator for the covariance matrix is a biased estimate of the covariance matrix even in the multi-variate case, and we need to apply Bessel's correction, i.e. use *n*-1 instead of *n* in the denominator, just like in the univariate case.

Multivariate Central Limit Theorem

- Consider independent and identically distributed vectors $\mathbf{X}_1$, $\mathbf{X}_2$, …, $\mathbf{X}_n$ in R^d drawn from a distribution with well-defined mean $\boldsymbol{\mu}$ and well-defined covariance matrix **C**.
- Then as $n \rightarrow \infty$, their mean sample mean has the following distribution:

$$\bar{\boldsymbol{X}}_{\boldsymbol{n}} := \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{X}_i$$

$$\sqrt{n}(\bar{\boldsymbol{X}}_{\boldsymbol{n}} - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{C})$$

Principal Components Analysis

- PCA has many applications – apart from face/object recognition – in image processing/computer vision, statistics, econometrics, finance, agriculture, and you name it!

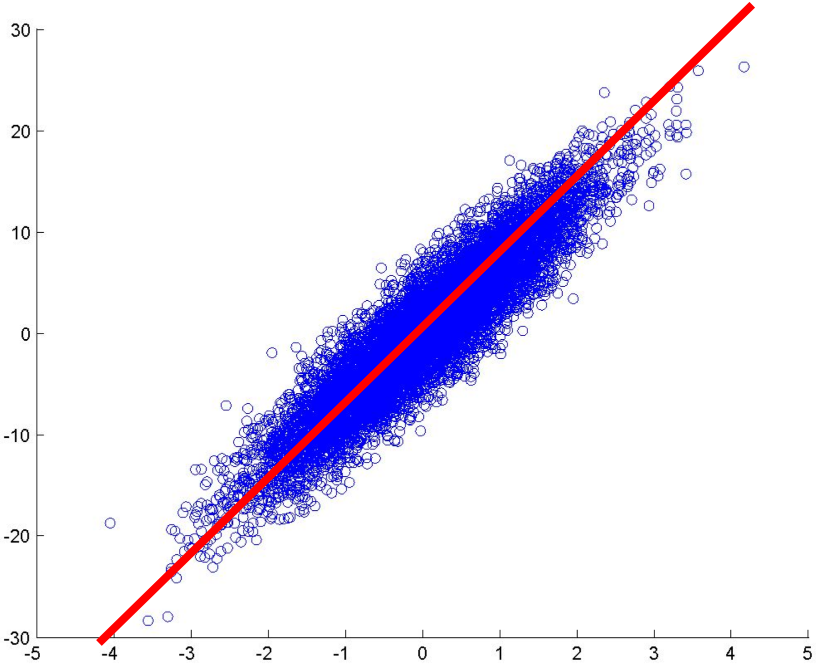
- Why PCA? What's special about PCA? See the next slides!

PCA: what does it do?

- It finds '*k*' perpendicular directions (all passing through the mean vector) such that the original data are approximated as **accurately** as possible when projected onto these '*k*' directions.

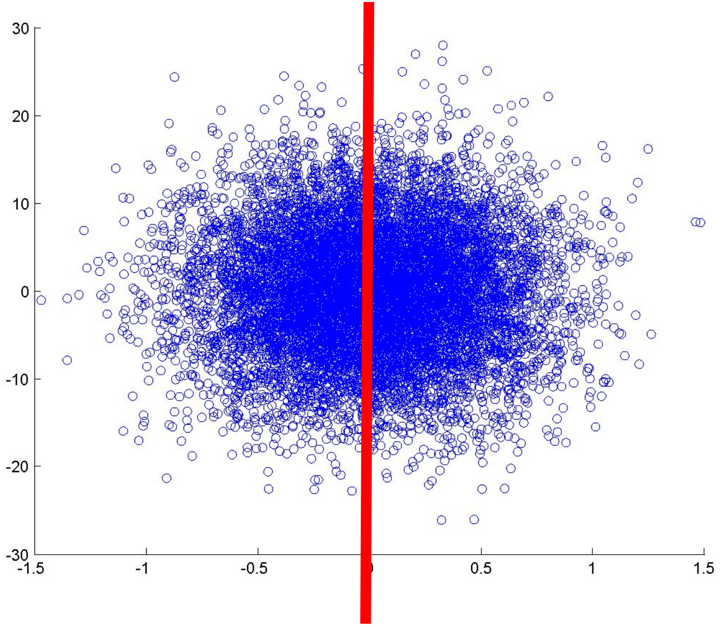
- We will see soon why these '*k*' directions are eigenvectors of the covariance matrix of the data!

PCA



Look at this scatter-plot of points in 2D. The points are highly spread out in the direction of the slanted red line.

PCA



- This is how the data would look if they were rotated in such a way that the major axis of the ellipse (the red line) now coincided with the Y axis.
- As the spread of the X coordinates is now relatively insignificant (<u>**observe the axes!**</u>), we can approximate the rotated data points by their projections onto the Y-axis (i.e. their Y coordinates alone!). This was not possible prior to rotation!

PCA

- As we could ignore the X-coordinates of the points post rotation and represent them just by the Y-coordinates, we have performed some sort of lossy data compression – or dimensionality reduction.

- The job of PCA is to perform such a rotation as shown on the previous two slides!

PCA

- **Aim of PCA:** Find the unit vector **u** passing through the sample mean (i.e. $\bar{\mathbf{x}}$), such that the projection of any mean-deducted point $\mathbf{x}_i - \bar{\mathbf{x}}$ onto **u**, most accurately approximates it.

$$\bar{\boldsymbol{x}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i.$$

Mean deducted point

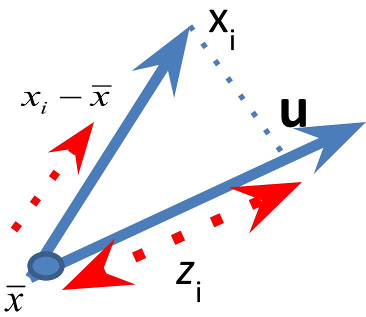
$$\boldsymbol{y}_i = \boldsymbol{x}_i - \bar{\boldsymbol{x}}, \qquad i = 1, \ldots, n.$$

$$z_i = \boldsymbol{u}^\top \boldsymbol{y}_i \qquad \hat{\boldsymbol{y}}_i = (\boldsymbol{u}^\top \boldsymbol{y}_i)\,\boldsymbol{u}.$$

Projection of y_i onto **u**

$$\|\boldsymbol{y}_i - \hat{\boldsymbol{y}}_i\|^2 = \|\boldsymbol{y}_i - (\boldsymbol{u}^\top \boldsymbol{y}_i)\boldsymbol{u}\|^2.$$

Projection error for y_i



$$\|\boldsymbol{u}\| = 1$$

$$\begin{aligned}\|\boldsymbol{y}_i - (\boldsymbol{u}^\top \boldsymbol{y}_i)\boldsymbol{u}\|^2 &= \boldsymbol{y}_i^\top \boldsymbol{y}_i - 2(\boldsymbol{u}^\top \boldsymbol{y}_i)(\boldsymbol{u}^\top \boldsymbol{y}_i) + (\boldsymbol{u}^\top \boldsymbol{y}_i)^2(\boldsymbol{u}^\top \boldsymbol{u}) \\ &= \boldsymbol{y}_i^\top \boldsymbol{y}_i - (\boldsymbol{u}^\top \boldsymbol{y}_i)^2.\end{aligned}$$

Total reconstruction error J(**u**)

$$J(\boldsymbol{u}) = \sum_{i=1}^{n} \|\boldsymbol{y}_i - \hat{\boldsymbol{y}}_i\|^2 = \sum_{i=1}^{n} \|\boldsymbol{y}_i\|^2 - \sum_{i=1}^{n} (\boldsymbol{u}^\top \boldsymbol{y}_i)^2$$

The first term does not depend on **u**, so minimizing J(**u**) is equivalent to maximizing the following:

$$\sum_{i=1}^{n} (\boldsymbol{u}^\top \boldsymbol{y}_i)^2.$$

$$\boldsymbol{S} = \sum_{i=1}^{n} \boldsymbol{y}_i \boldsymbol{y}_i^\top .$$

Scatter matrix (proportional to the sample covariance matrix)

$$\sum_{i=1}^{n} (\boldsymbol{u}^\top \boldsymbol{y}_i)^2 = \boldsymbol{u}^\top \boldsymbol{S} \boldsymbol{u}$$

we must maximize $\boldsymbol{u}^\top \boldsymbol{S} \boldsymbol{u}$ subject to $\|\boldsymbol{u}\| = 1$

$$\boldsymbol{S} = \boldsymbol{V} \boldsymbol{\Lambda} \boldsymbol{V}^\top$$

As **S** is symmetric, **V** is orthonormal and eigenvalues in $\boldsymbol{\Lambda}$ are non-negative

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$$

$$\boldsymbol{u} = \sum_{j=1}^{d} a_j \boldsymbol{v}_j, \qquad \text{so} \quad \sum_{j=1}^{d} a_j^2 = 1$$

Express **u** as a linear combination of the eigenvectors – this is possible as **V** is an orthonormal basis. That is any vector in R^d can be expressed as a linear combination of the columns of **V**

$$\boldsymbol{u}^\top \boldsymbol{S}\boldsymbol{u} = \sum_{j=1}^{d} \lambda_j a_j^2. \qquad \sum_{j=1}^{d} a_j^2 = 1 \qquad \boldsymbol{u} = \sum_{j=1}^{d} a_j \boldsymbol{v}_j$$

It is easy to show this if you express **u** = **Va** where a is a d x 1 vector

- The quantity $\mathbf{u}^\top\mathbf{S}\mathbf{u}$ is maximized if $a_1 = 1$ and the other a_j values are all zero.
- Note that $\lambda_1 \geqslant \lambda_2 \geqslant \ldots \geqslant \lambda_d$ – then we will have $\mathbf{u} = \mathbf{v}_1$ which is the eigenvector corresponding to the largest eigenvalue λ_1.

- Thus the direction **u** such that projection onto it yields the least total projection error is given by the eigenvector of the scatter matrix **S** (equivalently the covariance matrix) with the **largest eigenvalue**.
- This direction also captures the **maximum variance** of the data.

PCA

- PCA thus projects the data onto that direction that minimizes the total squared difference between the data-points and their respective projections along that direction.

- This **equivalently** yields the direction along which the spread (or variance) will be maximum.

- Why? Note that the eigenvalue of a covariance matrix tells you the variance of the data when projected along that particular eigenvector:

$$\boldsymbol{S}\boldsymbol{u} = \lambda \boldsymbol{u}$$

$$\boldsymbol{u}^\top \boldsymbol{S}\boldsymbol{u} = \lambda = \sum_{i=1}^{n} (\boldsymbol{u}^\top (\boldsymbol{x}_i - \bar{\boldsymbol{x}})^2$$

This term is proportional to the variance of the data when projected along **u**.

PCA

- But for most applications (including face recognition), just a single direction is absolutely insufficient!
- We will need to project the data (from the high-dimensional, i.e. d-dimensional space) onto k ($k << d$) different mutually perpendicular directions.
- What is the criterion for deriving these directions?
- We seek those k directions for which the total reconstruction error of all the N images when projected on those directions is minimized.

PCA

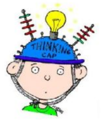
- We seek those *k* directions for which the total reconstruction error of all the *N* images when projected on those directions is minimized.

$$J(\{\boldsymbol{u}_j\}_{j=1}^k) := \sum_{i=1}^n \|\boldsymbol{y}_i - \sum_{j=1}^k (\boldsymbol{u}_j)^\top \boldsymbol{y}_i \boldsymbol{u}_j\|_2^2$$

- One can prove that these *k* directions will be the eigenvectors of the **S** matrix (equivalently covariance matrix of the data) corresponding to the *k*-largest eigenvalues. These *k* directions form the eigen-space.

PCA

- One can prove that these *k* directions will be the eigenvectors of the **S** matrix (equivalently covariance matrix of the data) corresponding to the *k*-largest eigenvalues. These *k* directions form the eigen-space.

- **Sketch of the proof:** 

✔ Assume we have found $\mathbf{u}_1$ and are looking for $\mathbf{u}_2$ (where $\mathbf{u}_2$ is perpendicular to $\mathbf{u}_1$ and $\mathbf{u}_2$ has unit magnitude).

✔ Write out the objective function with the two constraints.

✔ Minimize it and do some algebra to see that $\mathbf{u}_2$ is the eigenvector of **S** with the second largest eigenvalue.

✔ Proceed similarly for other directions.

How to pick *k* in an actual application?

- Trial and error. Usually between 50 to 100 for images of size 200 x 200.
- Divide your training set into two parts – *A* and *B* (B is usually called the **validation set**). Pick the value of *k* that gives the best recognition rate on B when you train on A.
- Stick to that value of *k*.
- Note: a larger *k* implies a better reconstruction but it may even cause a *decrease* in the recognition accuracy!

How to pick *k* in an actual application?

- Note: a larger *k* implies a better reconstruction but it may even cause a *decrease* in the recognition accuracy!

- Why – because throwing out some of the eigenvectors may lead to filtering of the data, removing some unnecessary artifacts for example.

Some observations about PCA for face images

- The matrix **V** (with all columns) is orthonormal. Hence the squared error between any image and its approximation using just top *k* eigenvectors is given by:

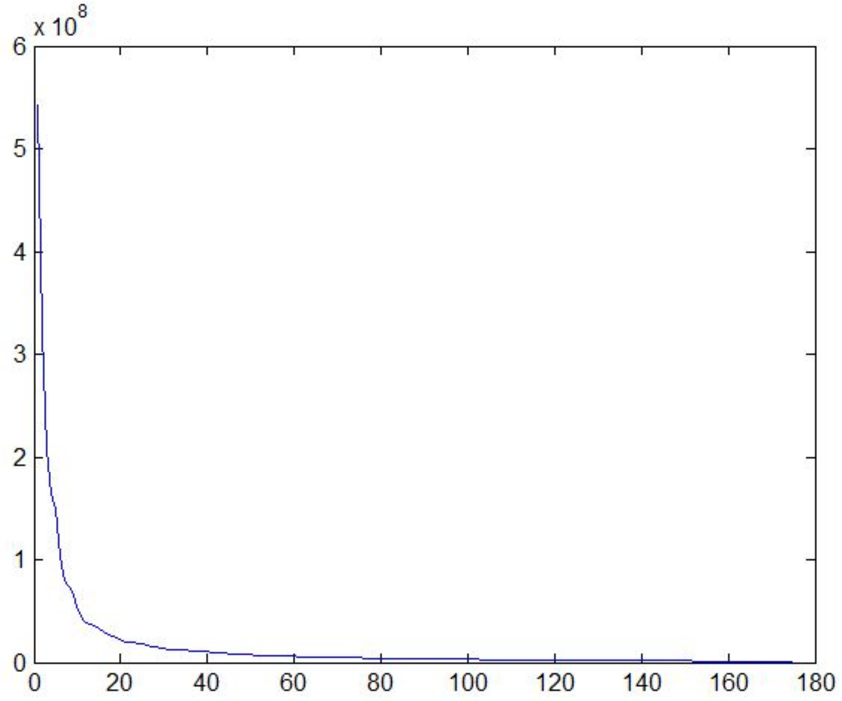
$$\|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 = \|V(\boldsymbol{\alpha}_i - \tilde{\boldsymbol{\alpha}}_i)\|^2 (why?)$$

$$= \|\boldsymbol{\alpha}_i - \tilde{\boldsymbol{\alpha}}_i\|^2 (why?)$$

$$= \sum_{j=k+1}^{d} \boldsymbol{\alpha}_{\mathrm{i}}^2 (why?)$$

Vector whose first *k* entries are identical to those in $\boldsymbol{\alpha}_i$, and the rest are 0.

This error is small ***on an average for a well-aligned group of face images*** – we will see why on the next slide.



X-axis = eigenvalue index (in sorted, descending order)
Y-axis = eigenvalue

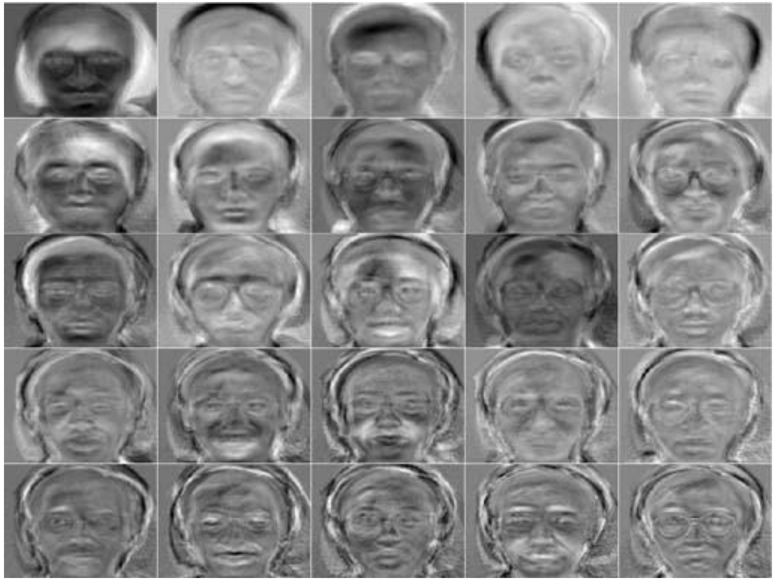
The eigenvalues of the covariance matrix typically decay fast in value (if the faces were properly normalized). Note that the *j*-th eigenvalue is proportional to the variance of the j-th eigencoefficient, i.e.

$$\lambda_j = \mathbf{e_j^t S e_j} = \sum_{i=1}^{N} \mathbf{e_j^t}(\mathbf{x_i} - \overline{\mathbf{x}})(\mathbf{x_i} - \overline{\mathbf{x}})^t \mathbf{e_j} = (N-1)E(\alpha_{ij}^2)$$

What this means is that the data have low variance when projected along most of the eigenvectors, i.e. effectively the data are concentrated in a lower-dimensional subspace of the d-dimensional space.

Example

The Yale Face database



Top 25 eigenfaces from the previous database: each eigenface is an eigenvector of the covariance matrix reshaped to form an image

Reconstruction of a face image using the top 1,8,16,32,…,104 eigenfaces (i.e. *k* varied from 1 to 104 in steps of 8)

$$\mathbf{x_i} = \bar{\mathbf{x}} + \hat{\mathbf{V}}\boldsymbol{\alpha}_\mathbf{i} \approx \bar{\mathbf{x}} + \sum_{l=1}^{k} \hat{\mathbf{V}}(:,l)\boldsymbol{\alpha}_\mathbf{ik}(l)$$